

JKBOSE 2026

CLASS 11TH

Bhaichara
ON
TOP

#XUV ♡

HYDROCARBONS

LECTURE-01

CHEMISTRY

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Topics To Covered

- 1 Isomerism**
- 2 Conformation of Ethane**

JKBOSE SYLLABUS

ORGANIC
CHEM $\Rightarrow$ $\frac{11}{+10}$
21



10 MARKS

➤ UNIT-X: HYDROCARBONS

- **Alkanes:** Nomenclature, isomerism, conformations (ethane ~~1,2~~ only), physical properties. Chemical reactions including free radical mechanism of halogenation, combustion and Pyrolysis.
- **Alkenes:** Nomenclature, structure of double bond (ethene), geometrical isomerism, methods of preparation, physical properties, chemical reactions- addition of hydrogen, halogen, water, hydrogen halides (Markownikov's addition and peroxide effect). ozonolysis, oxidation, mechanism of electrophilic addition.
- **Alkynes:** Nomenclature, structure of triple bond (ethyne), physical properties, methods of preparation, chemical reactions: acidic character of alkynes, addition reaction of hydrogen, halogens, hydrogen halides and water, **Aromatic hydrocarbons** introduction, IUPAC nomenclature, Benzene resonance, aromaticity, chemical properties, mechanism of electrophilic substitution-nitration, sulphonation, halogenations Friedel Craft's alkylation and acylation, directive influence of functional group in mono substituted benzene.

PART-I

ALKANES

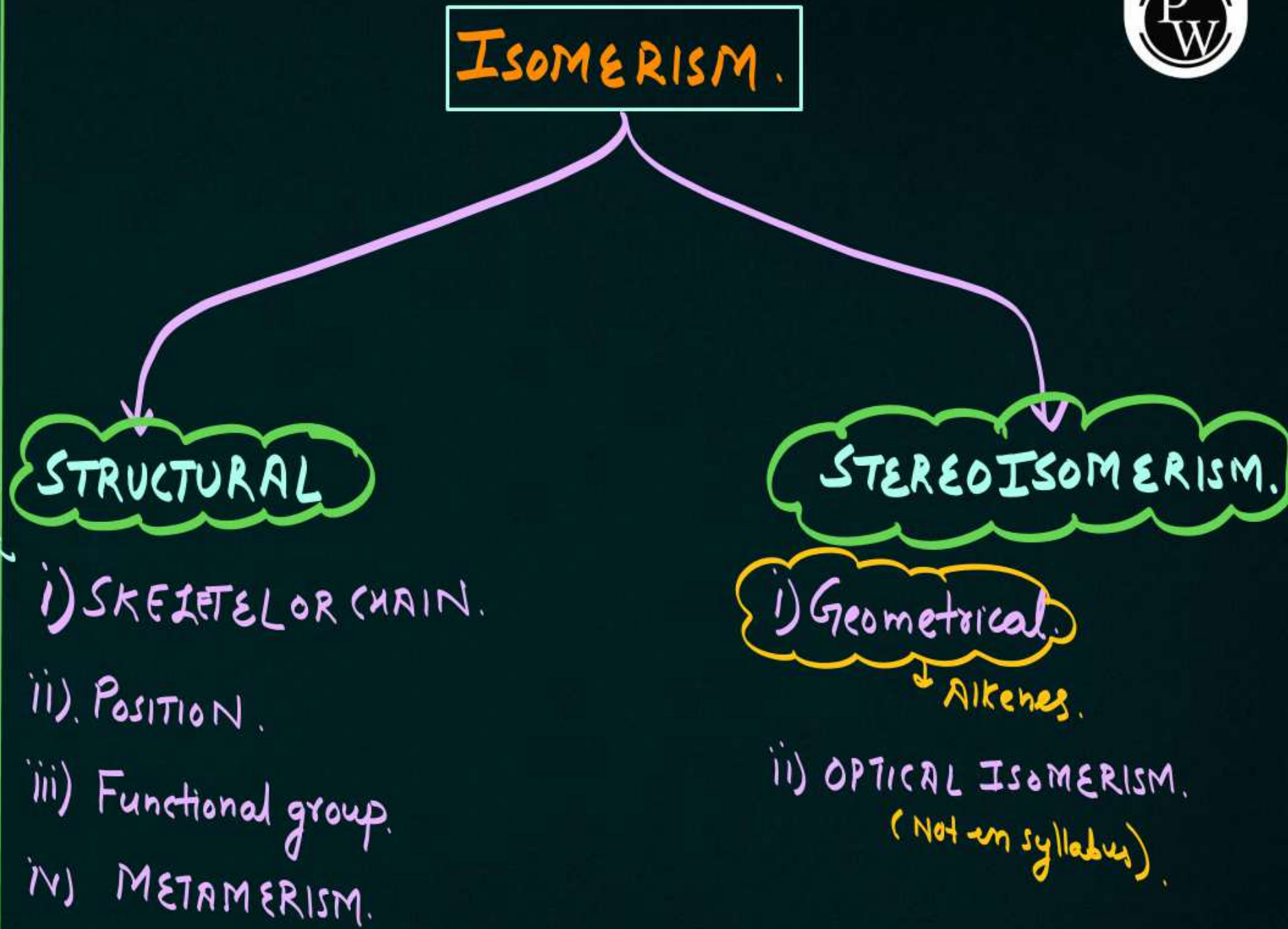
ISOMERISM.

Different molecules with same molecular formula but different physical & chemical properties are called isomers. This whole phenomena is known as Isomerism.

• ISOMERISM is OF Two Types.

a) STRUCTURAL ISOMERISM.

b) STEREOISOMERISM.



a) STRUCTURAL ISOMERISM

Molecules have same molecular formula but different structure is called as structural isomers. This phenomena is called as structural isomerism.

Four types

i) Skeletal or Chain isomerism

ii) Position isomerism

iii) Functional group isomerism

iv) Metamerism.

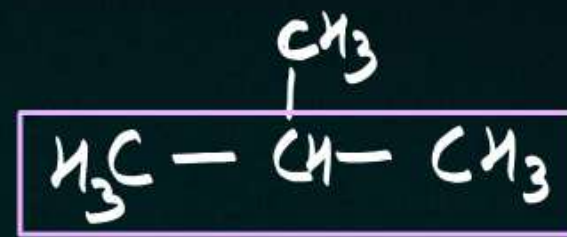
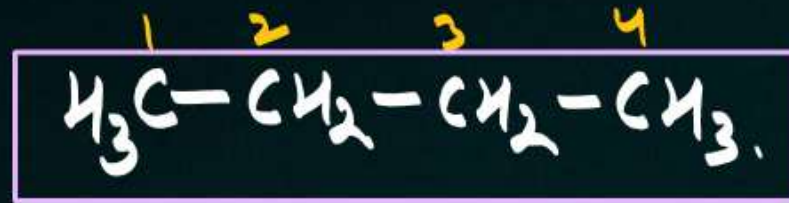
i) SKELETAL OR CHAIN ISOMERISM.



Molecules having same molecular formula but different carbon chain within the molecule are called chain isomers and phenomena is known as Chain isomerism.

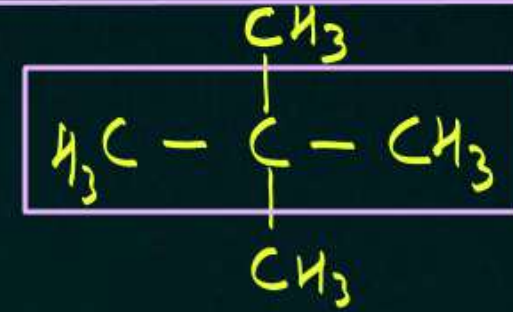
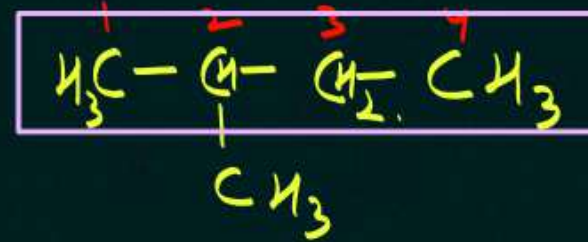
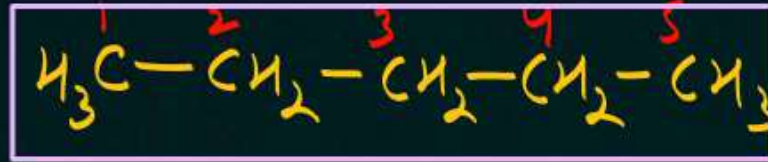
- First three members of alkane (Methane, Ethane & Propane) do not show this isomerism.

Butane (C_4H_{10})



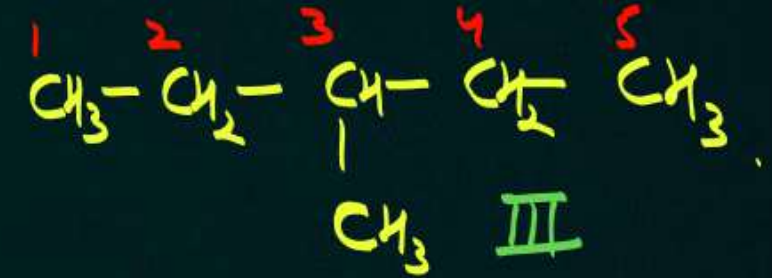
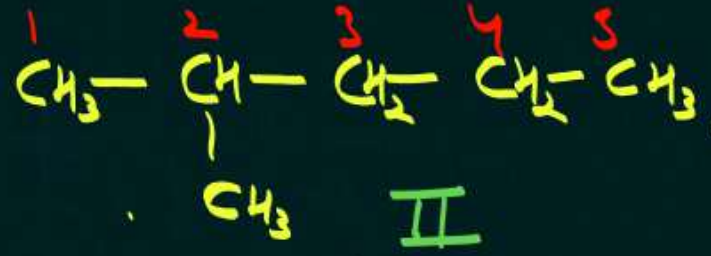
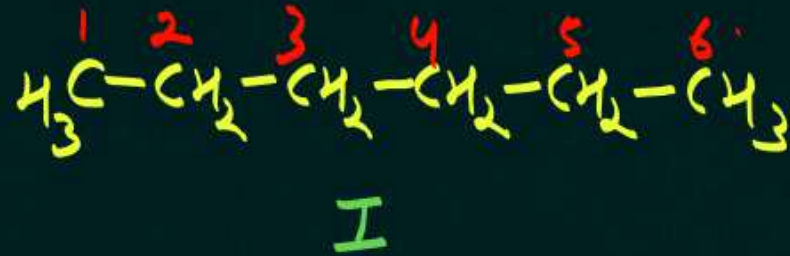
No. of chain isomers = Two.

Pentane (C_5H_{12})

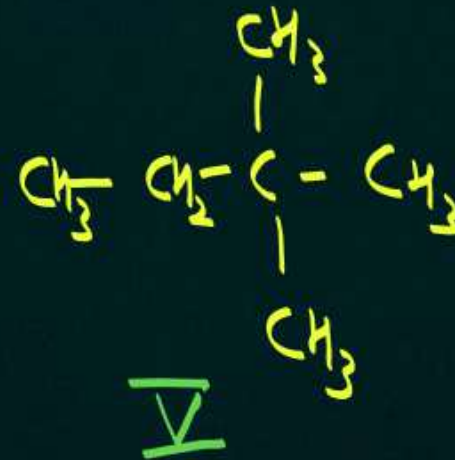


No. of chain isomers = 3

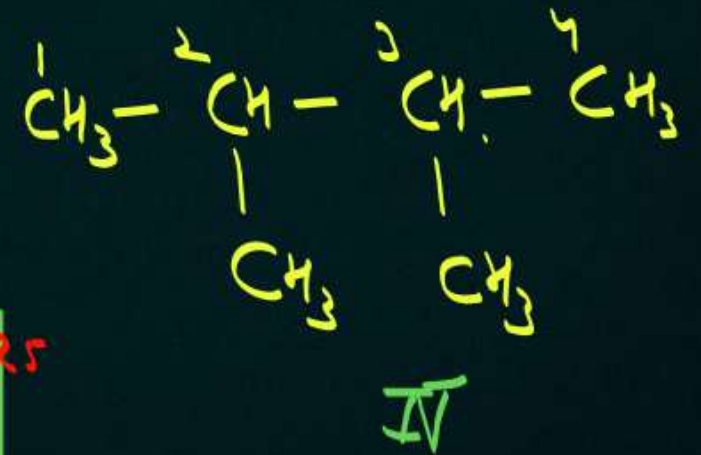
HEXANE (C_6H_{14})



HEPTANE (C_7H_{16}) = No. of chain isomers = 9



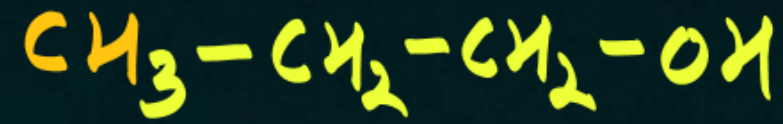
No. of chain isomers = 5



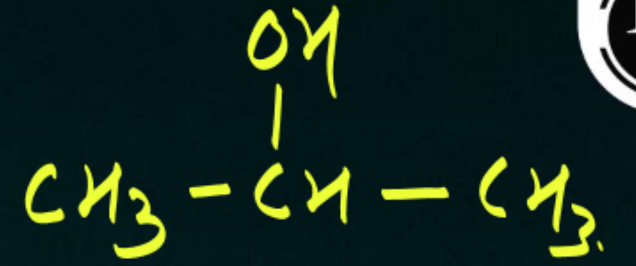
ii) POSITION ISOMERISM. $\Rightarrow$

Molecules having same molecular formula but different position of same functional group, multiple bond, or substituent are called POSITION ISOMERS, and phenomena is called as position isomerism.

e.g 1)



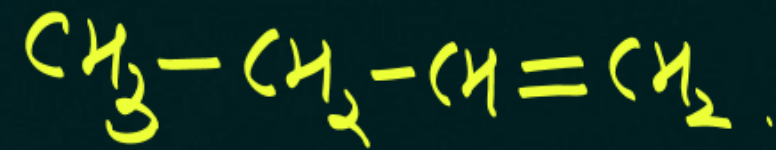
Propanol.



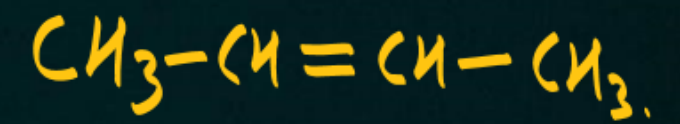
Propan-2-ol.

(molecular formula = $\text{C}_3\text{H}_8\text{O}$)

e.g. ②



But-1-ene.



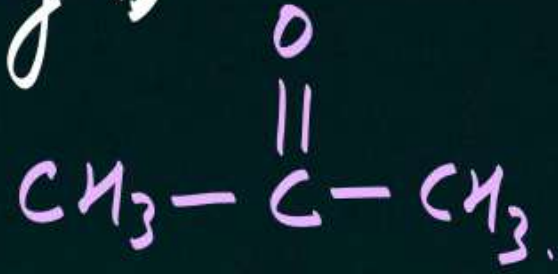
But-2-ene

(molecular formula. C_4H_8)

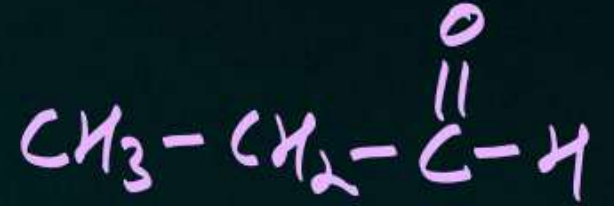
C) Functional group Isomerism

The compounds having same molecular formula but different functional group are called functional group isomers. and isomerism is called as functional group isomerism.

e.g 1)



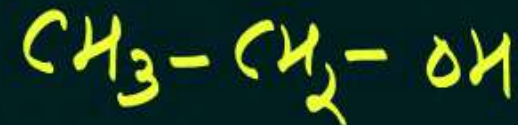
Propanone



Propanal.

(Molecular formula
 $\text{C}_3\text{H}_6\text{O}$)

e.g (2)



Ethanol



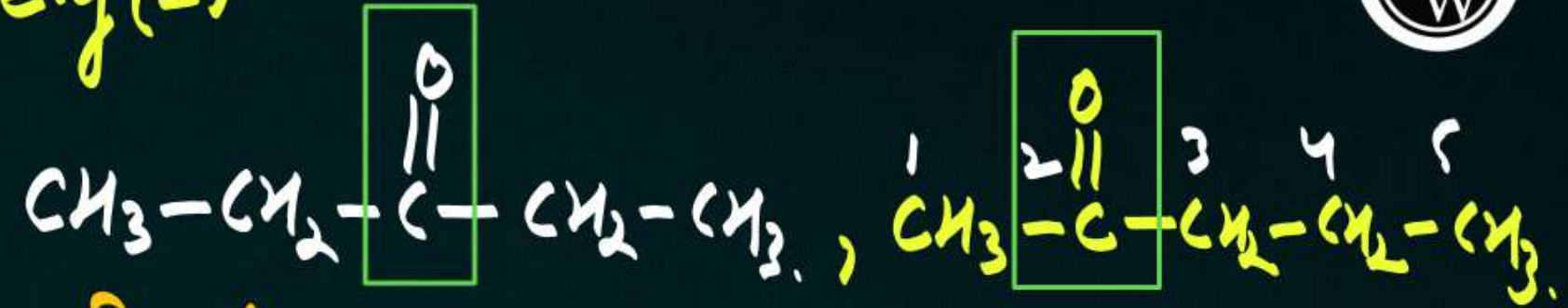
Methoxymethane

(Molecular formula
 $\text{C}_2\text{H}_6\text{O}$)

IV) METAMERISM

Compounds having same molecular formula, same functional group but different no. of carbon atoms on both side of functional group are called Metamers and isomerism is known as Metamerism.

e.g (1)



Pentan-3-one.

Pentan-2-one

(Molecular formula = $\text{C}_5\text{H}_{10}\text{O}$)

e.g (2)



Methoxypropane

Ethoxyethane.

(Molecular formula = $\text{C}_4\text{H}_{10}\text{O}$)

QUESTION

Eclipsed and the staggered conformations can be represented by

- 1 Sawhorse projection
- 2 Newman projection
- 3 Both (1) and (2)
- 4 None of the above

QUESTION



How many chain isomers can be formed by butane?

- 1 One
- 2 Three
- 3 Two
- 4 Four

QUESTION



The given structures,



1 Conformational isomers

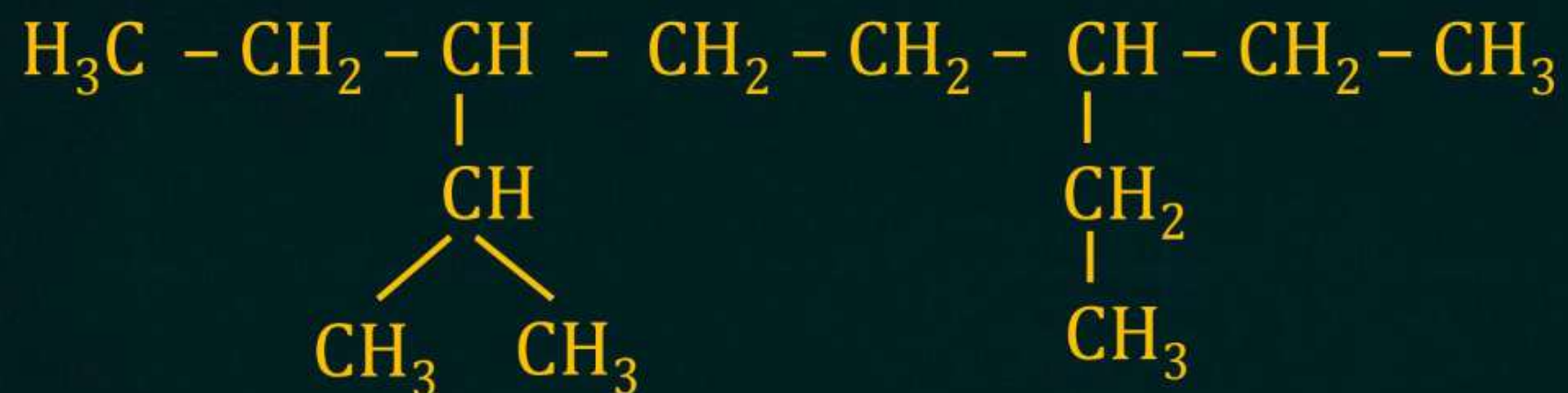
2 Chain isomers

3 Position isomers

4 Functional isomers

QUESTION

The correct IUPAC name of the following alkane is



- | | |
|-------------------------------------|-------------------------------------|
| 1 3,6-diethyl-2-methyloctane | 2 5-isopropyl -3-ethyloctane |
| 3 3-ethyl-5-isopropyloctane | 4 3-isopropyl-6-ethyloctane |

Thank
You